**Experiment No: 1**

**Title: To create a Python script that scrapes data from a static website**

SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

KJSSE/AIDS/TYBTECH /SEM-V/WSAPI/2026-27

**Roll No.: 16014225801** **Experiment No.: 1**

**Aim:** To understand the basics of web scraping by extracting information from a static website using Python, Requests, and BeautifulSoup.

---

**Resources needed:**

- Python 3.x
- Visual Studio Code / PyCharm / Jupyter Notebook
- Internet Connection
- Python Libraries:
  - requests
  - beautifulsoup4
- Website: **https://books.toscrape.com** or any static website of your choice

---

Web scraping is the process of automatically extracting data from websites using software programs instead of manually copying information. It is widely used in market research, product price monitoring, news aggregation, data analysis, and AI applications.

A static website contains data directly in the HTML page returned by the server. Since the required information is already present in the HTML source code, it can be easily extracted using libraries such as Requests and BeautifulSoup.

The Requests library downloads the HTML content of a webpage, while BeautifulSoup parses the HTML document into a structured tree, allowing users to locate and extract specific elements using HTML tags and attributes.

**1. Static vs. Dynamic Websites:**

- Static Websites: Serve pre-rendered HTML content, where the data you see in your browser's "view page source" is directly present in the HTML document returned by the server.
- Dynamic Websites: Leverage JavaScript to load and/or render content in the browser, meaning the initial HTML might be a barebones structure, and data is fetched and displayed dynamically as a user interacts or the page loads further.
- Why the Difference Matters for Scraping: Static sites are easier to scrape because the content is readily available in the initial HTML, enabling the use of simpler tools. Dynamic sites require browser automation to execute JavaScript and interact with the page to retrieve the full content, making the process more complex.

**2. HTML Structure and Element Identification:**

- Web pages are built using HTML (HyperText Markup Language), which defines the structure and content with elements and tags.

- To effectively scrape a website, it's crucial to analyze its HTML structure using browser developer tools. This helps identify the specific HTML elements (like divs, paragraphs, links) and their attributes (like class names or IDs) that contain the target data. For example, the quote text on a page might be inside a <p> tag, which is nested within a  tag with a specific class.

**List of Activities:**

Students must perform the following activities during the laboratory session.

**Activity 1: Open the Target Website**

- Open **https://books.toscrape.com**
- Observe the webpage layout.
- Identify different information displayed for each book.

**Activity 2: Inspect HTML Structure**

Using **Browser Developer Tools**,

- Identify the HTML tag containing the page title.
- Identify the HTML tag containing book details.
- Identify the HTML tag containing book price.
- Identify the HTML tag containing book rating.
- Identify the HTML tag containing availability.

Record your observations.

**Activity 3: Install Required Libraries**

Install the required Python libraries.

pip install requests

pip install beautifulsoup4

**Activity 4: Download HTML Source Code**

Write a Python program to

- Import Requests library.
- Download webpage HTML.
- Check HTTP Status Code.
- Display first 500 characters of HTML.

**Activity 5: Parse HTML**

Create a BeautifulSoup object using

BeautifulSoup(response.text,"html.parser")

Display

- Page Title
- Main Heading

**Activity 6: Extract Single Elements**
Using find(), extract

- Page Title
- First Heading
- First Paragraph
- First Hyperlink

**Activity 7: Extract Multiple Elements**
Using find_all()
Display

- All paragraph tags
- All hyperlinks
- Total number of paragraph tags

K. J. SOMAIYA COLLEGE OF ENGG.

**Activity 8: Extract Data Using HTML Attributes**
Extract

- First Book Price
- First Book Rating
- First Book Image URL
- First Book Link

using HTML attributes (class, href, src).

**Activity 9: Extract Details of All Books**
Using
find_all("article", class_="product_pod")
extract for every book

- Book Title
- Price
- Rating
- Availability

Display the extracted information in a formatted manner.

**Activity 10: Observation**
Record

- Number of books extracted
- HTML tags used
- Attributes used
- Functions used (find(), find_all())

**Implementation:**

```
import requests
from bs4 import BeautifulSoup

# ----------------------------------------
# Activity 4: Download HTML Source Code
# ----------------------------------------

url = "https://books.toscrape.com"

response = requests.get(url)

# ----------------------------------------
# Activity 5: Parse HTML
# ----------------------------------------

soup = BeautifulSoup(response.text, "html.parser")
main_heading = soup.find("h1")

# ----------------------------------------
# Activity 6: Extract Single Elements
# ----------------------------------------

print("\n" + "=" * 60)
print("Activity 6: Single Elements")

page_title = soup.find("title").text
print("Page Title:", page_title)

first_heading = soup.find("h1").text
print("First Heading:", first_heading)

first_paragraph = soup.find("p").text
print("First Paragraph:", first_paragraph)

first_link = soup.find("a")["href"]
print("First Hyperlink:", first_link)

# ----------------------------------------
# Activity 7: Extract Multiple Elements
# ----------------------------------------

print("\n" + "=" * 60)
print("Activity 7: Multiple Elements")

paragraphs = soup.find_all("p")

print("\nAll Paragraph Tags:\n")
for p in paragraphs:
    print(p.text.strip())

links = soup.find_all("a")
```

```
print("\nAll Hyperlinks:\n")
for link in links:
    print(link.get("href"))

print("\nTotal Paragraph Tags:", len(paragraphs))

# ----------------------------------------
# Activity 8: Extract Data Using Attributes
# ----------------------------------------

print("\n" + "=" * 60)
print("Activity 8: First Book Details")

first_book = soup.find("article", class_="product_pod")

title = first_book.h3.a["title"]

price = first_book.find("p", class_="price_color").text

rating = first_book.find("p", class_="star-rating")["class"][1]

availability = first_book.find("p", class_="instock availability").text.strip()

image_url = url + "/" + first_book.find("img")["src"].replace("../", "")

book_link = url + "/" + first_book.h3.a["href"]

print("Book Title :", title)
print("Price      :", price)
print("Rating     :", rating)
print("Availability:", availability)
print("Image URL  :", image_url)
print("Book Link  :", book_link)

# ----------------------------------------
# Activity 9: Extract Details of All Books
# ----------------------------------------

print("\n" + "=" * 60)
print("Activity 9: Details of All Books")

books = soup.find_all("article", class_="product_pod")

count = 0

for book in books:
    count += 1

    title = book.h3.a["title"]

    price = book.find("p", class_="price_color").text

    rating = book.find("p", class_="star-rating")["class"][1]
```

```
    availability = book.find("p", class_="instock availability").text.strip()

    print("-" * 50)
    print("Book", count)
    print("Title        :", title)
    print("Price        :", price)
    print("Rating       :", rating)
    print("Availability :", availability)

# ----------------------------------------
# Activity 10: Observation
# ----------------------------------------

print("\n" + "=" * 60)
print("Activity 10: Observation")

print("Number of Books Extracted :", len(books))

print("\nHTML Tags Used:")
print("1. <title>")
print("2. <h1>")
print("3. <p>")
print("4. <a>")
print("5. <article>")
print("6. <img>")

print("\nAttributes Used:")
print("1. class")
print("2. href")
print("3. src")
print("4. title")

print("\nFunctions Used:")
print("1. find()")
print("2. find_all()")

print("=" * 60)
```

**Results: Paste screenshots.**

```
============================================================
Activity 7: Multiple Elements
All Paragraph Tags:

Â£51.77
In stock

Â£53.74
In stock

Â£50.10
In stock
```

```
Â£47.82
In stock

Â£54.23
In stock

Â£22.65
In stock

Â£33.34
In stock

Â£17.93
In stock

Â£22.60
In stock

Â£52.15
In stock

Â£13.99
In stock

Â£20.66
In stock

Â£17.46
In stock

Â£52.29
In stock

Â£35.02
In stock

Â£57.25
In stock

Â£23.88
In stock

Â£37.59
In stock

Â£51.33
In stock

Â£45.17
In stock

All Hyperlinks:
```

```
index.html
index.html
catalogue/category/books_1/index.html
catalogue/category/books/travel_2/index.html
catalogue/category/books/mystery_3/index.html
catalogue/category/books/historical-fiction_4/index.html
catalogue/category/books/sequential-art_5/index.html
catalogue/category/books/classics_6/index.html
catalogue/category/books/philosophy_7/index.html
catalogue/category/books/romance_8/index.html
catalogue/category/books/womens-fiction_9/index.html
catalogue/category/books/fiction_10/index.html
catalogue/category/books/childrens_11/index.html
catalogue/category/books/religion_12/index.html
catalogue/category/books/nonfiction_13/index.html
catalogue/category/books/music_14/index.html
catalogue/category/books/default_15/index.html
catalogue/category/books/science-fiction_16/index.html
catalogue/category/books/sports-and-games_17/index.html
catalogue/category/books/add-a-comment_18/index.html
catalogue/category/books/fantasy_19/index.html
catalogue/category/books/new-adult_20/index.html
catalogue/category/books/young-adult_21/index.html
catalogue/category/books/science_22/index.html
catalogue/category/books/poetry_23/index.html
catalogue/category/books/paranormal_24/index.html
catalogue/category/books/art_25/index.html
catalogue/category/books/psychology_26/index.html
catalogue/category/books/autobiography_27/index.html
catalogue/category/books/parenting_28/index.html
catalogue/category/books/adult-fiction_29/index.html
catalogue/category/books/humor_30/index.html
catalogue/category/books/horror_31/index.html
catalogue/category/books/history_32/index.html
catalogue/category/books/food-and-drink_33/index.html
catalogue/category/books/christian-fiction_34/index.html
catalogue/category/books/business_35/index.html
catalogue/category/books/biography_36/index.html
catalogue/category/books/thriller_37/index.html
catalogue/category/books/contemporary_38/index.html
catalogue/category/books/spirituality_39/index.html
catalogue/category/books/academic_40/index.html
catalogue/category/books/self-help_41/index.html
catalogue/category/books/historical_42/index.html
catalogue/category/books/christian_43/index.html
catalogue/category/books/suspense_44/index.html
catalogue/category/books/short-stories_45/index.html
catalogue/category/books/novels_46/index.html
catalogue/category/books/health_47/index.html
catalogue/category/books/politics_48/index.html
catalogue/category/books/cultural_49/index.html
catalogue/category/books/erotica_50/index.html
catalogue/category/books/crime_51/index.html
```

```
catalogue/a-light-in-the-attic_1000/index.html
catalogue/a-light-in-the-attic_1000/index.html
catalogue/tipping-the-velvet_999/index.html
catalogue/tipping-the-velvet_999/index.html
catalogue/soumission_998/index.html
catalogue/soumission_998/index.html
catalogue/sharp-objects_997/index.html
catalogue/sharp-objects_997/index.html
catalogue/sapiens-a-brief-history-of-humankind_996/index.html
catalogue/sapiens-a-brief-history-of-humankind_996/index.html
catalogue/the-requiem-red_995/index.html
catalogue/the-requiem-red_995/index.html
catalogue/the-dirty-little-secrets-of-getting-your-dream-job_994/index.html
catalogue/the-dirty-little-secrets-of-getting-your-dream-job_994/index.html
catalogue/the-coming-woman-a-novel-based-on-the-life-of-the-infamous-feminist-
victoria-woodhull_993/index.html
catalogue/the-coming-woman-a-novel-based-on-the-life-of-the-infamous-feminist-
victoria-woodhull_993/index.html
catalogue/the-boys-in-the-boat-nine-americans-and-their-epic-quest-for-gold-at-the-
1936-berlin-olympics_992/index.html
catalogue/the-boys-in-the-boat-nine-americans-and-their-epic-quest-for-gold-at-the-
1936-berlin-olympics_992/index.html
catalogue/the-black-maria_991/index.html
catalogue/the-black-maria_991/index.html
catalogue/starving-hearts-triangular-trade-trilogy-1_990/index.html
catalogue/starving-hearts-triangular-trade-trilogy-1_990/index.html
catalogue/shakespeares-sonnets_989/index.html
catalogue/shakespeares-sonnets_989/index.html
catalogue/set-me-free_988/index.html
catalogue/set-me-free_988/index.html
catalogue/scott-pilgrims-precious-little-life-scott-pilgrim-1_987/index.html
catalogue/scott-pilgrims-precious-little-life-scott-pilgrim-1_987/index.html
catalogue/rip-it-up-and-start-again_986/index.html
catalogue/rip-it-up-and-start-again_986/index.html
catalogue/our-band-could-be-your-life-scenes-from-the-american-indie-underground-
1981-1991_985/index.html
catalogue/our-band-could-be-your-life-scenes-from-the-american-indie-underground-
1981-1991_985/index.html
catalogue/olio_984/index.html
catalogue/olio_984/index.html
catalogue/mesaerion-the-best-science-fiction-stories-1800-1849_983/index.html
catalogue/mesaerion-the-best-science-fiction-stories-1800-1849_983/index.html
catalogue/libertarianism-for-beginners_982/index.html
catalogue/libertarianism-for-beginners_982/index.html
catalogue/its-only-the-himalayas_981/index.html
catalogue/its-only-the-himalayas_981/index.html
catalogue/page-2.html

Total Paragraph Tags: 60

============================================================
Activity 8: First Book Details
Book Title : A Light in the Attic
```

```
Price      : Â£51.77
Rating     : Three
Availability: In stock
Image URL  :
https://books.toscrape.com/media/cache/2c/da/2cdad67c44b002e7ead0cc35693c0e8b.jpg
Book Link  : https://books.toscrape.com/catalogue/a-light-in-the-
attic_1000/index.html

============================================================
Activity 9: Details of All Books
------------------------------------------------
Book 1
Title        : A Light in the Attic
Price        : Â£51.77
Rating       : Three
Availability : In stock
------------------------------------------------
Book 2
Title        : Tipping the Velvet
Price        : Â£53.74
Rating       : One
Availability : In stock
------------------------------------------------
Book 3
Title        : Soumission
Price        : Â£50.10
Rating       : One
Availability : In stock
------------------------------------------------
Book 4
Title        : Sharp Objects
Price        : Â£47.82
Rating       : Four
Availability : In stock
------------------------------------------------
Book 5
Title        : Sapiens: A Brief History of Humankind
Price        : Â£54.23
Rating       : Five
Availability : In stock
------------------------------------------------
Book 6
Title        : The Requiem Red
Price        : Â£22.65
Rating       : One
Availability : In stock
------------------------------------------------
Book 7
Title        : The Dirty Little Secrets of Getting Your Dream Job
Price        : Â£33.34
Rating       : Four
Availability : In stock
------------------------------------------------
```

```
Book 8
Title        : The Coming Woman: A Novel Based on the Life of the Infamous Feminist,
Victoria Woodhull
Price        : Â£17.93
Rating       : Three
Availability : In stock
--------------------------------------------------
Book 9
Title        : The Boys in the Boat: Nine Americans and Their Epic Quest for Gold at
the 1936 Berlin Olympics
Price        : Â£22.60
Rating       : Four
Availability : In stock
--------------------------------------------------
Book 10
Title        : The Black Maria
Price        : Â£52.15
Rating       : One
Availability : In stock
--------------------------------------------------
Book 11
Title        : Starving Hearts (Triangular Trade Trilogy, #1)
Price        : Â£13.99
Rating       : Two
Availability : In stock
--------------------------------------------------
Book 12
Title        : Shakespeare's Sonnets
Price        : Â£20.66
Rating       : Four
Availability : In stock
--------------------------------------------------
Book 13
Title        : Set Me Free
Price        : Â£17.46
Rating       : Five
Availability : In stock
--------------------------------------------------
Book 14
Title        : Scott Pilgrim's Precious Little Life (Scott Pilgrim #1)
Price        : Â£52.29
Rating       : Five
Availability : In stock
--------------------------------------------------
Book 15
Title        : Rip it Up and Start Again
Price        : Â£35.02
Rating       : Five
Availability : In stock
--------------------------------------------------
Book 16
Title        : Our Band Could Be Your Life: Scenes from the American Indie
Underground, 1981–1991
```

```
Price        : Â£57.25
Rating       : Three
Availability : In stock
--------------------------------------------------
Book 17
Title        : Olio
Price        : Â£23.88
Rating       : One
Availability : In stock
--------------------------------------------------
Book 18
Title        : Mesaerion: The Best Science Fiction Stories 1800–1849
Price        : Â£37.59
Rating       : One
Availability : In stock
--------------------------------------------------
Book 19
Title        : Libertarianism for Beginners
Price        : Â£51.33
Rating       : Two
Availability : In stock
--------------------------------------------------
Book 20
Title        : It's Only the Himalayas
Price        : Â£45.17
Rating       : Two
Availability : In stock



============================================================
Activity 10: Observation
Number of Books Extracted : 20

HTML Tags Used:
1. <title>
2. <h1>
3. <p>
4. <a>
5. <article>
6. <img>

Attributes Used:
1. class
2. href
3. src
4. title

Functions Used:
1. find()
2. find_all()
============================================================
```

**Post-Lab Questions:**

A government weather website publishes district-wise daily heatwave bulletins as a static webpage. Describe how you would use Requests and BeautifulSoup to extract the district name, maximum temperature, and heatwave alert level. Which HTML tags or attributes would you inspect before writing the scraper?

**Answer**

To extract district-wise heatwave information from a government weather website, I would first use the **Requests** library to download the webpage by sending an HTTP GET request. After confirming that the request is successful (status code 200), I would parse the HTML using **BeautifulSoup**. Before writing the scraper, I would inspect the webpage using the browser's Developer Tools to identify the HTML structure. Specifically, I would look for the tags containing the district name (such as <td>, , or ), the maximum temperature, and the heatwave alert level. I would also note important attributes such as class or id that uniquely identify these elements. Using BeautifulSoup's find() and find_all() methods, I could then extract the required information and display or store it in a structured format such as a list or CSV file.

---

**Conclusion:**

This experiment demonstrated how to use Python, Requests, and BeautifulSoup to scrape data from a static website efficiently. HTML tags and attributes were analyzed to extract structured information accurately. The practical implementation improved understanding of web scraping techniques, HTML parsing, and automated data extraction for real-world applications.

**Grade: AA / AB / BB / BC / CC / CD /DD**

**Signature of faculty in-charge with date**

---

**References:**

**Books/ Journals/ Websites:**

1. Seppe vanden Broucke, Bart Baesens , “Practical Web Scraping for Data Science Best Practices and Examples with Python”, Apress, 3rd Edition 2018.
2. Anish Chapagain, “Hands-On Web Scraping with Python Extract Quality Data from the Web Using Effective Python Techniques”, Packt Publishing, 2nd Edition 2023.
3. Pradumna Milind Panditrao, “A Python Guide for Web Scraping: Explore Python Tools, Web Scraping Technique”, BPB, 2nd Edition 2019.